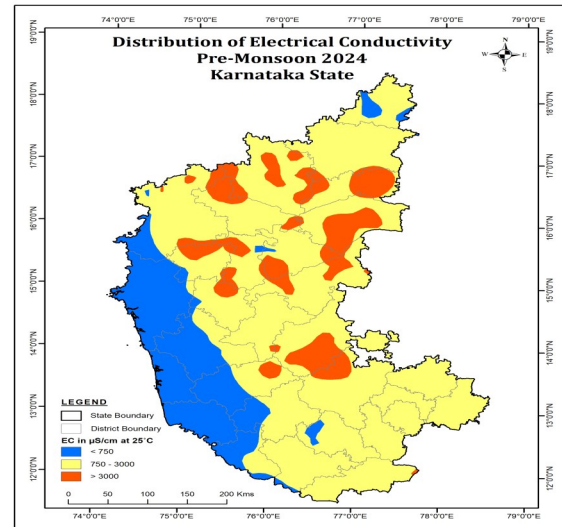
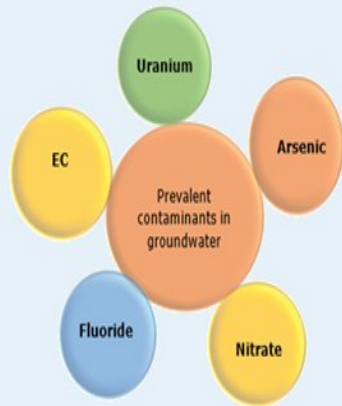


Groundwater Quality Scenario in Karnataka

Parameters	No of samples	Permissible limit	No. of Samples above permissible limit	% Samples above permissible Limit
EC	398	3000 $\mu\text{S/cm}$	60	15.08
Fluoride	398	1.5 mg/L	67	16.83
Nitrate	398	45 mg/L	181	45.48
Arsenic	118	10 ppb	2	1.69
Uranium	118	30 ppb	10	8.47



Districts with anomalous values at sporadic locations

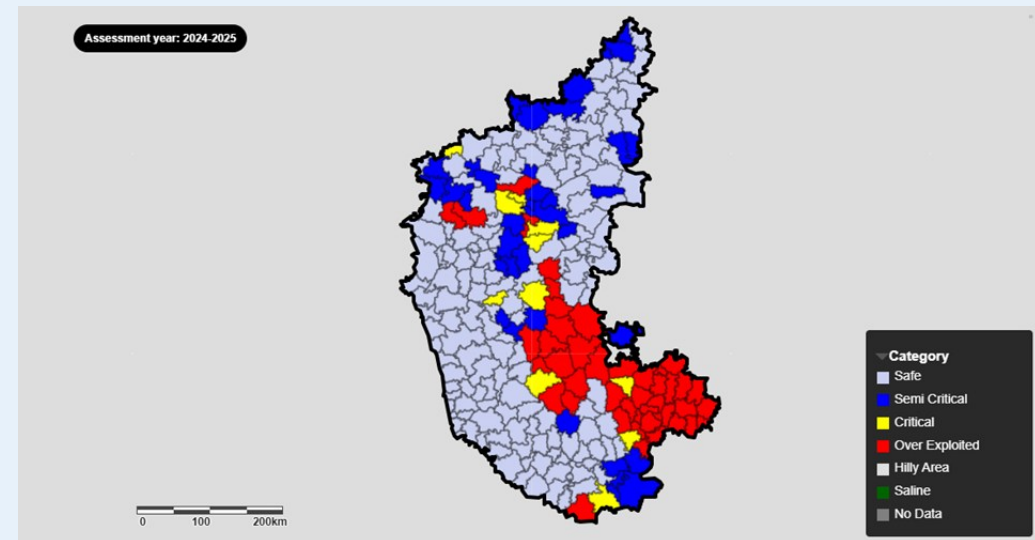
EC (3000 $\mu\text{S/cm}$)	Bagalkot, Belagavi, Belgaum, Bijapur, Chikkamagaluru, Chitradurga, Dharwad, Gadag, Koppal, Raichur, Tumkur, Vijayanagara, Vijayapura, Yadgiri
Fluoride (F > 1.5 mg/L)	Bagalkot, Bellary, Bengaluru Rural, Chikkaballapura, Chitradurga, Davanagere, Dharwad, Gadag, Haveri, Kalaburagi, Kolar, Koppal, Mysuru, Raichur, Ramanagara, Tumkur, Vijayapura, Yadgiri
Nitrate (Nitrate > 45 mg/L)	Bagalkot, Belagavi, Belgaum, Bellary, Bengaluru Rural, Bengaluru Urban, Bidar, Bijapur, Chamarajanagara, Chikkaballapura, Chitradurga, Dakshina Kannada, Davanagere, Dharwad, Gadag, Gulbarga, Hassan, Haveri, Kalaburagi, Kodagu, Kolar, Koppal, Mandya, Mysuru, Raichur, Ramanagara, Shivmoga, Tumkur, Vijayanagara, Vijayapura, Yadgiri
Arsenic (As > 10 ppb)	Gadag, Raichur

For Further Information, Contact to :
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Email: chmn-cgwb@nic.in

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Central Ground Water Board
Department of Water Resources, RD & GR
Ministry of Jal Shakti, Government of India



**Dynamic Ground Water Resources, Ground Water Level &
Ground Water Quality of Karnataka, 2025**

November, 2025

Groundwater Resource Scenario in Karnataka

- ◆ Ground Water Resources Assessment (GWRA)- jointly carried out by Central Ground Water Board and State Nodal/Ground Water Department periodically as per the Ground Water Resource Estimation Committee (GEC) methodology.
- ◆ Carried out under the guidance of the respective State/UT Level Committees (SLCs) and overall supervision of Central Level Expert Group (CLEG).
- ◆ As part of the assessment, 'Annual Extractable Ground Water Resource' as well as 'Annual Ground Water Extraction' are assessed for each assessment unit (Taluk).
- ◆ The 'Stage of Ground Water Extraction' is computed as the ratio of 'Annual Ground Water Extraction' with respect to 'Annual Extractable Ground Water Resource' and is usually expressed in percentage. Based on the stage of extraction, the assessment units are categorized as Safe ($\leq 70\%$), Semi-Critical ($>70\%$ and $\leq 90\%$), Critical ($>90\%$ and $\leq 100\%$) and Over-Exploited ($>100\%$).
- ◆ GWRA-2025, 2024, 2023, 2022 and 2020 has been carried out through a software/web-based application "INDIA-GROUNDWATER RESOURCE ESTIMATION SYSTEM (IN-GRES)" developed by CGWB through IIT-Hyderabad.

Salient Features

1	Rainfall	1,035.07 mm
2	Hydrogeology	Major parts of the State are occupied by hardrock of Archean age. Northern parts area occupied by Deccan basalts. Recent alluvium is restricted to a narrow belt in coastal areas and along stream courses.
3	Recharge Worthy Area of the State	1.71 Lakh Sq. Km
4	Assessment Unit (AU) Type / Number	Taluk / 237 Numbers
5	Average area of Assessment Unit	721.52 Sq. Km

Findings

	Attribute	GWRA-2017	GWRA-2020	GWRA-2022	GWRA-2023	GWRA-2024	GWRA-2025
1	Total Annual Ground Water Recharge (in bcm)	16.84	18.16	17.74	18.93	18.74	19.27
2	Annual Extractable Ground Water Resources (in bcm)	14.79	16.4	16.04	17.08	16.88	17.41
3	Annual Ground Water Extraction (in bcm)	10.34	10.63	11.22	11.32	11.55	11.58
4	Stage of Ground Water Extraction (in %)	69.87	64.85	69.93	66.26	68.44	66.49

bcm: Billion Cubic Meters

Categorization of Assessment Units based on the 'Stage of Ground Water Extraction'

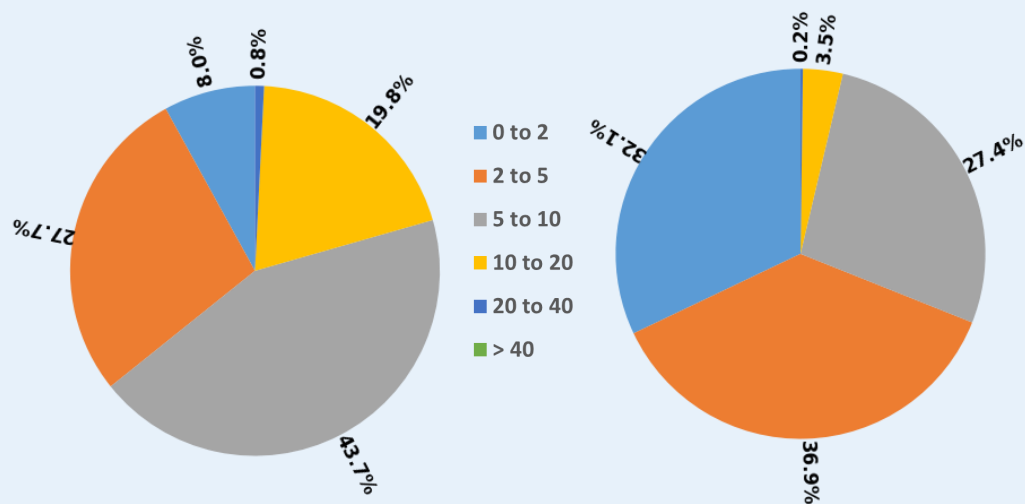
S. No	Category	GWRA-2017		GWRA-2020		GWRA-2022		GWRA-2023		GWRA-2024		GWRA-2025	
		Number of AUs	% of AUs	Number of AUs	% of AUs	Number of AUs	% of AUs	Number of AUs	% of AUs	Number of AUs	% of AUs	Number of AUs	% of AUs
1	Safe	97	55	130	57	139	59	146	62.39	144	60.76	145	61.18
2	Semi-critical	26	15	35	15	35	15	32	13.68	33	13.92	36	15.19
3	Critical	8	5	10	4	11	5	12	5.13	15	6.33	11	4.64
4	Over-exploited	45	26	52	23	49	21	44	18.8	45	18.99	45	18.99
5	Saline												
Total number of AUs		176		227		234		234		237		237	

Recommendations: -

- * Major portion of the State is covered by Peninsular Gneisses, Granites and Dharwar Schists of Archaean age. Substantial area in the northern part of Karnataka is underlain by basalts, which form a continuation of the Deccan Traps occurring in Maharashtra. The sedimentary formations comprising Bhima and Kaladgis occupy a small area in the northern districts. The recent alluvium is restricted to a narrow belt in the coastal area and along stream courses. The aquifer systems are classified into nine major groups depending upon their characteristics and are Banded Gneissic Complex (BGC), Basalt, Schists, Granites, Charnockites, Limestones, Laterites, Sandstones and alluvium.
- * the Annual Ground Water Recharge has been assessed as 19.27 bcm and the Annual Extractable Ground Water resource is 17.41 bcm. The present Annual Ground Water Extraction is 11.58 bcm and the Stage of Ground Water Extraction is 66.49%. Out of the 237 assessment units (taluks), 45 units (18.99%) have been categorized as 'Over exploited', 11 units (4.64%) as 'Critical', and 36 units (15.19 %) as 'Semi critical' and 145 units (61.18 %) have been categorized as 'Safe'. There is no taluk under "Saline" category.
- * More numbers of Water Harvesting and Conservation Structures may be constructed to catch the rain as the State is blessed with more than 1000 mm annual rainfall particularly in the hard rock terrain. State may also effectively use "Master plan for Artificial Recharge" prepared by CGWB in consultation with State Government (<https://cgwb.gov.in/cgwbpm/publication-detail/324>).
- * Restoration /rejuvenation of all the existing tanks should be taken up with the view of accommodating the available surface run off and thus augmentation of the ground water resources by artificial recharge. Periodical maintenance of these tanks is to be ensured. The "Manual on Artificial Recharge Techniques for augmentation of ground water" prepared by CGWB may be used for planning. (<https://cgwb.gov.in/sites/default/files/MainLinks/Manual-Artificial-Recharge.pdf>).
- * National Aquifer Mapping & Management Programme (NAQIM) Reports prepared by CGWB (<https://cgwb.gov.in/cgwbpm>) which are also being shared with State/District Authorities and Ground Water Year Book published by CGWB having water level & water quality data may be used in Ground water management. (<https://cgwb.gov.in/cgwbpm>).
- * Irrigation sector is consuming about 89% of ground water resources, hence micro-irrigation techniques needs to be adopted in more areas to increase the irrigation efficiency and reduce the load on ground water.
- * Creating awareness (Mass Awareness Campaign for public and farmers, slideshows, display boards on water conservation, Water Management Training Programme for personnel related with water sector, painting/essay competition for school students etc.) regarding water conservation etc may be organized at appropriate level.
- * In urban and industrial areas, disposal of industrial effluents, solid waste and urban sewerage should be disposed off safely after treatment, so that the phreatic aquifer does not get adversely polluted.
- * Regulation & control of Ground water Extraction: Ministry of Jal Shakti has issued the guidelines for control and regulations of ground water extraction vide notification dated 24.09.2020 which has further been amended in March 2023. Concerned departments may ensure implementations of the guidelines.

Groundwater Level Scenario in Karnataka

1. Depth to Groundwater Level

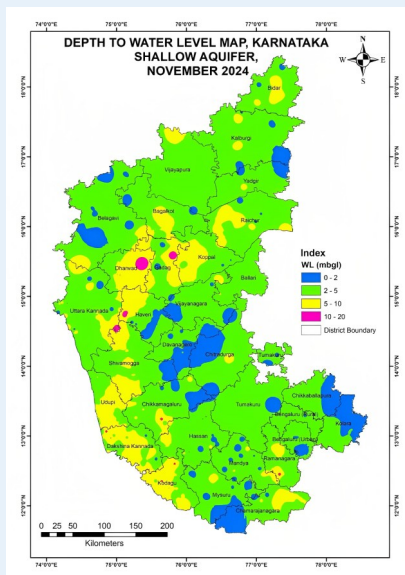
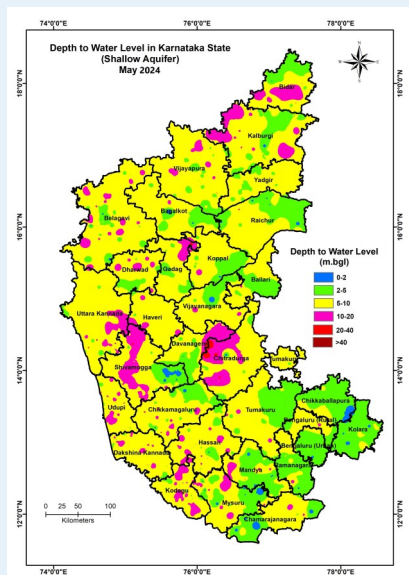


Pre-Monsoon (May 2024)

About 79% of wells recorded groundwater levels within 0–10 m bgl, 7.8% in < 2 m bgl, and only 0.78% in > 20 m bgl, with overall depths indicating generally favorable groundwater conditions.

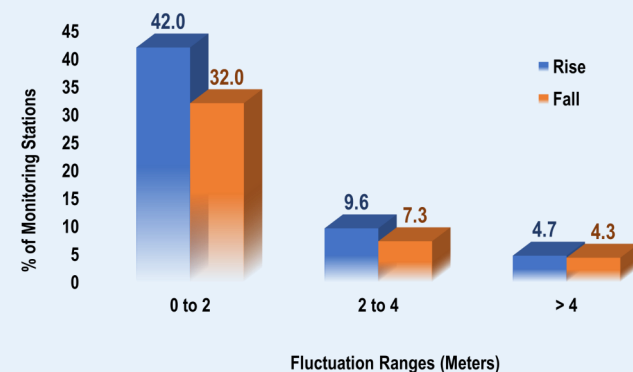
Post-Monsoon (November 2024)

About 96% of wells recorded groundwater levels within 0–10 m bgl, 31% in < 2 m bgl, and < 0.2% in > 20 m bgl, indicating generally favorable post-monsoon groundwater conditions.



2. Decadal Fluctuation of Groundwater Levels

Pre-Monsoon (May 2024 vs Decadal Mean: May 2014–2023)



Post-Monsoon (November 2024 vs Decadal Mean: November 2014–2023)

